



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - Standard investigation

Drafted 30/09/2026

INVESTIGATION WORKFLOW SUITE

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Executive summary

What happened

At 02:47 local on 11/08/2026, an empty 220 t class rigid haul truck, HT-407, descending the main ramp of the Stage 4 open pit at Bindarra Gold Mine struck the right side of light vehicle LV-118 as it emerged from the RL 312 dewatering spur. The truck operator braked from 37 km/h, and impact was at 24 km/h. LV-118 was pushed 18.4 m down the ramp and rolled onto its left side against the inner windrow.

Its driver was a grade control technician with a survey subcontractor, on his third night shift at the mine. He sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. He remained conscious, was extricated 27 minutes after the collision and was flown to the metropolitan trauma centre at 07:40. His seatbelt was fastened and loaded. The roll-over protection structure deformed but was not breached.

LV-118 was destroyed. HT-407 was damaged at the front left and returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes, and the ramp below RL 340 was closed for 22 hours.

Why it happened

It was extremely likely that neither operator was aware of the other until about four seconds remained, and this extremely likely contributed to the collision. The following conditions produced this lack of awareness:

- **Geometry.** The spur met the ramp 34 m beyond a crest. A descending operator had 41 m of sight line, against a stopping distance of 96 m and a plan minimum of 120 m. There was no sign, give-way control, delineation or posted speed. This very likely contributed.
- **Spur construction.** The spur was opened under a works order classified as routine earthworks, so no management of change record was raised. That record was the only route to updating the traffic management plan, the in-cab road network and the collision awareness system configuration. None was updated. This very likely contributed.
- **Radio calls.** Positive calls were required only at nine listed intersections. The list could change only at the annual review, and nothing required new roads to be referred to the plan's owner. This very likely contributed. Neither operator made a call, which the plan and local practice allowed. This likely contributed.
- **Ramp entry.** LV-118 entered at about 25 km/h without stopping or establishing the truck's position. This very likely contributed, although a full stop would have added only a few metres of sight line.
- **Induction.** It was extremely likely that the driver's contractor induction was accepted as equivalent without verifying the site supplement or pit driving assessment. This likely contributed, although he already applied the rule it would have taught, that light vehicles give way at all times.
- **Collision awareness.** A loading-area geofence extended northward in June 2026 suppressed alerts at the intersection. This likely contributed. An alert would have come after braking began, but months of silence on that ramp had shaped the operator's expectation that silence meant an empty road.
- **Guidance.** Guideline MSG-14 leaves the review trigger and the choice of interaction zones to each site. This likely contributed, on balance and with less confidence than the other factors. The mine was not in breach, and this is not a finding against the Inspectorate.

The truck operator braked and steered towards the wall side within 1.4 seconds of sighting LV-118. This reduced the impact speed and kept LV-118 away from the drop.

A wet ramp surface and degraded truck brakes were ruled out. Whether dust reduced the operator's view could not be resolved.

Two conditions increased risk without contributing. The operator was on night 10 of a 14-night roster, in the circadian low. Night shift extrication depended on a second responder travelling from the village, which delayed it by approximately 9 minutes.

Safety issues

Six safety issues were identified:

- the suppression geofence
- the undefined terms in MSG-14
- the works order exclusion from management of change
- the annually fixed intersection list
- unverified induction equivalence
- the night shift emergency response arrangement, which did not contribute

Safety action

At 05:10 on the night of the occurrence, the Mining Manager directed that a positive call be made before entering any Stage 4 intersection, listed or not, until further notice. [NOT ESTABLISHED: safety actions taken or proposed by the operator, the contractors or the Inspectorate, and any recommendations made, in response to each safety issue]

Event overview

Times are local, eight hours ahead of UTC; seconds are retained where recorded data gives them.

The RL 312 dewatering spur

The RL 312 dewatering spur is a 240 m access road in the Stage 4 open pit at Bindarra Gold Mine. It was cut under works order WO-26-4471 to reach dewatering bore DW-14, and the works were completed on 31/07/2026. On 02/08/2026 the bore was energised, light vehicles first used the spur, and the ROAD CLOSED - CONSTRUCTION sign at the spur mouth was removed and not replaced. From then until the occurrence the intersection had no warning sign on either ramp approach, no give-way or stop control, no delineation and no posted speed for the spur.

The spur joins the main ramp from the east, on the inside of a left-hand bend, at an included angle of 71 degrees, 34 m beyond the high point of a crest. There the ramp descends at 9.1 per cent and is 24 m wide between windrows. The survey found that a descending haul truck operator first had line of sight to the spur mouth at 41 m, and a driver at the spur mouth first had line of sight to a descending truck at 38 m, the 1.6 m inner windrow hiding the truck's lower body before that.

An earlier near miss at the same intersection

At about 22:10 on 04/08/2026, a dewatering technician came out of the spur in front of a descending haul truck he estimated was about 60 m away, reversed back and let it pass. That night he submitted hazard report HAZ-2026-0917, rated Medium, writing that "you cannot see up the ramp from that road", that there was no sign warning the trucks and no give-way on the spur side, and that "I do not think they know the road is there."

The Dewatering Superintendent closed the report on 06/08/2026, recording that the crew had been reminded at the 05/08 pre-start to take care and give way to haul trucks, and reduced the rating to Low. The report was not referred to the mining department, the contractor, the survey department or the owner of the traffic management plan.

The night of 11/08/2026: people and conditions

The haul truck operator was driving HT-407, a 220 t class rigid haul truck, on night 10 of a 14-night swing and 10 hours 47 minutes into a 12 hour 30 minute shift. He had slept 6 hours 30 minutes in the preceding day and had 7 years' experience on the truck type and 14 months at Bindarra. He described himself as tired in the way operators are on night ten, but awake and driving properly.

The light vehicle driver was a grade control technician employed by a survey subcontractor, on his third night shift at Bindarra. His induction was a two-hour online general mining module and a 20 minute gatehouse briefing before his site access card was issued on 09/08/2026. He said he had driven the spur six or seven times over the previous week without meeting a haul truck at the intersection. No permit, call-in or board existed by which the shift supervisor or dispatch could have known his vehicle, LV-118, was in the pit, and the night shift supervisor did not know it was there.

At 02:22:41 dispatch called the Stage 4 units back to work after crib. HT-407 completed one load, dumped at 02:38:55, and its operator called that he was coming back down. The fleet management system recorded the empty truck at 41 km/h at RL 368 at 02:45:58, and 39 km/h at RL 348 at 02:46:31 with the retarder in the second position.

The in-cab road network, last published on 24/07/2026, showed no intersection at RL 312. The operator said he knew the spur had been cut and had noticed the closed sign was gone, but believed the spur was still closed.

The light vehicle leaves the bore

After finishing a set of levels at the DW-14 collar, the driver drove back towards the ramp. His departure time was not established precisely; he recalled leaving at about ten to three, later than the recorded collision time. He said he slowed at the spur mouth, looked left up the ramp, saw nothing and drove out. He first said he had stopped, then corrected himself: he did not believe he came to a complete stop. He estimated his speed at about 25 km/h. He said he could see a piece of road before the rise before the windrow cut off the view, and had assumed he would see a truck's lights on the wall first. His radio was on and turned up, but he did not remember hearing anything on it.

Neither driver made a positive radio call. The RL 312 spur intersection was not one of the nine intersections at which the traffic management plan requires one.

Sighting and collision

At 02:47:06, the last steady data point, the truck was at RL 314, 41 m above the intersection, at 37 km/h. The operator said he saw the light vehicle at about 40 m, matching the surveyed sight line to within a metre. At 02:47:07.4, 1.4 seconds later, service brake pressure rose from 0 to 620 kPa in 0.4 seconds, the retarder went to maximum and 14 degrees of right steering was applied. He said he braked and steered right into the windrow in the same movement; to the left was a drop of about 100 m, with the light vehicle ahead.

At 02:47:09.2, 1.8 seconds after braking began, the truck had slowed to 32 km/h and its collision awareness system logged a proximity event with LV-118 at 21.4 m, closing at 13 km/h. Because the intersection lay inside the S4-LOAD suppression geofence, nothing was presented in the cab.

At 02:47:12.0 the truck's front left struck the right side of LV-118 at 24 km/h, 6.2 m into the ramp from the spur mouth and 3.1 m out from the inner windrow toe. Mean deceleration from brake application to impact was 1.85 m/s^2 , against 1.9 m/s^2 in the unit's brake test on 03/08/2026. The light vehicle was pushed 18.4 m down the ramp; at 11.2 m its tyre marks became a body panel gouge where it went over onto its left side against the inner windrow. HT-407 was stationary at 02:47:14.3, park brake applied, 19 m below the intersection.

Conditions at the time

The ramp surface was dry and well maintained. The truck operator said some dust was hanging but could not say whether it made a difference; the light vehicle driver described the air as clear and said he could see the ramp and windrow without difficulty. LV-118's beacon was on; its whip was sheared in the collision and its lamp worked when tested afterwards.

Immediate aftermath

At 02:47:41 the truck operator reported the collision and requested the emergency response team. At 02:48:01, 49 seconds after the collision, dispatch stopped all Stage 4 units, holding the operator in his cab until the ramp was clear, then releasing him at 02:48:44. The emergency response team was paged at 02:48. At 02:49:30 the operator reported that the driver was conscious and talking but could not get out because his door was against the ground. The supervisor, travelling from the ROM pad, directed that he was not to be moved.

Time	Event
02:51	First response vehicle entered the pit
02:54	First response vehicle reached the scene
03:07	Second responder arrived from the accommodation village, 11 km from the gatehouse, with the hydraulic rescue tool (a second responder is required before any extrication)
03:14	Driver extricated through the rear of the vehicle, 27 minutes after the collision
03:41	Driver reached the surface medical facility
07:40	Driver flown to the metropolitan trauma centre

The post-incident review found extrication was delayed by approximately 9 minutes waiting for the second responder and tool, without affecting the clinical outcome because the injuries were stable and the driver was conscious throughout.

The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. His seatbelt was found fastened, with loading marks on the webbing. The roll-over protection structure was deformed inward at the right rear mounting but not breached. LV-118 was destroyed. HT-407 was damaged at the front left bumper, left mudguard and lower access steps, and returned to service on 14/08/2026.

The haul truck operator and shift supervisor were stood down for the rest of the shift and tested for drugs and alcohol, both negative. The Stage 4 pit stood down for 9 hours 15 minutes and the main ramp below RL 340 was closed for 22 hours. At 05:10 the Mining Manager directed that a positive call be made on the pit channel before entering any Stage 4 intersection, listed or not, until further notice.

Analysis

Introduction

This analysis follows the chain of conditions that led a haul truck and a light vehicle to meet at the RL 312 spur intersection, each unaware of the other until about four seconds remained. Five separate lines of evidence lead to that one condition: the geometry of the intersection, the silence of the collision awareness system, the absence of any radio call, the light vehicle's entry onto the ramp, and the dust that may or may not have been present. The organisational arrangements that produced them are traced back to a works order, a traffic management plan whose list of controlled intersections could change only once a year, and a regulatory guideline that leaves the key terms to each site. Two further conditions, the night shift roster and the night shift emergency response arrangement, are analysed as factors that increased risk without contributing to this occurrence.

Times are local.

People

The light vehicle's entry onto the ramp

The light vehicle driver was a grade control technician on his third night shift at Bindarra. He had finished a set of levels at the DW-14 collar and drove the 240 m spur back towards the main ramp. His account was that he slowed at the spur mouth, looked left up the ramp, saw nothing and drove out at about 25 km/h. He first said he had stopped. When his account was read back to him, he corrected it to say that he did not believe he had come to a complete stop. That correction was against his own interest, and it agrees with the scene evidence: the impact point was 6.2 m into the ramp from the eastern edge of the spur mouth, and the gouge runs from there.

BIN-TMP-004 clause 4.1 provides that "Light vehicles shall give way to heavy vehicles at all times and in all circumstances." The driver described himself as applying that rule. He looked for a truck, saw none, and entered. What he could see from the spur mouth was limited by the geometry, analysed under Environment below. At his eye height of 1.45 m, the survey gives 38 m of sight line to a descending truck. The 1.6 m inner windrow hid the truck's lower body for the last 25 m of the spur's approach. A look left from a stationary position would therefore have shown him approximately what his look at 25 km/h showed him. He said he had expected to see a truck's lights on the wall first.

Nothing at the spur mouth told him to do anything different. There was no give-way control, no stop line, no warning sign and no posted speed. Clause 6.4 of the plan applies only to intersections shown on the road layout drawing, and this intersection was not on it. He had not completed the Bindarra light vehicle pit driving assessment and had never seen the traffic management plan. That gap is analysed under Organisation.

The evidence weighs against this action in two directions. Without the light vehicle on the ramp there would have been no collision, and entering without establishing the truck's position was the step that placed it there. Against that, the survey shows that a full stop would have gained the driver only a few metres of sight line, nowhere near the 96 m the truck needed to stop. The action was the proximate step. It was not one the driver could have made safe by care alone.

The investigation found it virtually certain that the light vehicle entered the main ramp from the spur without stopping and without establishing the haul truck's position, and very likely that this contributed to the collision.

No positive radio call from either vehicle

The pit channel was logged continuously. The investigator listened to the whole period, from the shift's return from crib to two minutes after the collision. Neither vehicle made any transmission about the RL 312 spur, the dewatering bore, or a light vehicle being anywhere in the Stage 4 pit. LV-118 made no transmission at all that shift after logging on at 01:04.

No call was required. Clause 4.3.2 of BIN-TMP-004 reads: "At intersections not listed at Appendix B, vehicles shall give way in accordance with clause 4.1. No call is required." The RL 312 intersection was not listed. The truck operator, the light vehicle driver and the night shift supervisor each described the same practice independently and without prompting: calls are made at the nine intersections on the crib room board and nowhere else. The supervisor said he had never corrected anybody for not calling elsewhere. He also said that requiring calls everywhere would make the channel unusable. The omission therefore belonged to no individual. The plan required no call at this intersection, and local practice matched the plan exactly.

A call would very probably have made a difference. The channel was open and both vehicles were on it. A call from either vehicle would have been heard by the other, by dispatch and by the supervisor. At 02:46:18 dispatch told the excavator operator that "407's about four minutes off you", so the dispatcher was tracking the truck closely enough to place it. The evidence against certainty is that a call gives knowledge, not separation. Even with each driver aware of the other, the two vehicles would still have had to decide who went first across 41 m of sight line.

The investigation found it virtually certain that neither operator made a positive call on the pit channel before entering the intersection, and likely that this contributed to the collision.

The truck operator's response

The fleet management system download records the truck's last steady data point at 02:47:06: 37 km/h, 41 m above the intersection. At 02:47:07.4 service brake pressure rose from 0 to 620 kPa in 0.4 seconds. In the same movement the retarder went to maximum and 14 degrees of right steering was applied. That is a response time of 1.4 seconds, inside the 1.5 seconds assumed in the stopping distance calculation used for the plan's sight distance requirement.

Mean deceleration from brake application to impact was 1.85 m/s^2 , against 1.9 m/s^2 in the unit's brake test on 03/08/2026. The truck struck the light vehicle at 24 km/h. The technical services analysis gives 37 km/h as the impact speed had no braking been applied. The braked scuff deviates 1.4 m to the right over its first 20 m, consistent with the recorded steering input. The operator's account was that to his left lay a drop of about 100 m, with the light vehicle ahead. He steered to the wall side, which was the only direction open to him.

After the impact he stopped 19 m below the intersection and reported the collision on the pit channel. He followed the dispatcher's instruction to stay in the cab until the ramp was stopped, then went down and stayed with the driver until the response team arrived.

The sight line meant a collision could not be avoided once the two vehicles were committed. The response affected how severe it was. Braking took a substantial part of the impact energy out of the collision, and the steering kept it against the wall side of the ramp. The difference between a 37 km/h and a 24 km/h impact on the side of a light vehicle separates the injuries actually sustained from a far more serious outcome.

The investigation found it virtually certain that the truck operator braked and steered to the right within 1.4 seconds of the first available sighting, and very likely that this reduced the severity of the outcome.

Fatigue

The truck operator was on night 10 of a 14-night swing, working 18:00 to 06:30. The collision occurred at 02:47, 10 hours 47 minutes into the shift, within the period of lowest circadian alertness. Every limit in the contractor's fatigue management standard was met:

- no more than 14 consecutive nights
- a minimum 10 hour break
- no shift extended beyond 12 hours 40 minutes in the preceding nine.

His voluntary sleep record shows 6 hours 30 minutes in the day before the occurrence and between 5 and 7 hours on each of the preceding nine days. In interview he described himself as tired in the way operators are on the tenth night, but not falling asleep, and awake and driving properly.

The case for fatigue having influenced the occurrence is that the eleventh hour of a tenth night shift is where degraded detection and slowed response are expected, and that detecting a hazard in an unlit pit is exactly the task fatigue degrades first.

The recorded data do not support it. The download gives 1.4 seconds from the last steady data point to a rise in brake pressure. That is inside the 1.5 second response assumed in the plan's stopping distance calculation and faster than the fleet figure. The braking and steering that followed were immediate, correct and in the only available direction. The arithmetic of the intersection also removes any role for response time: at 41 m of sight line and 37 km/h, a response time of zero still ends in a collision. There is no slowed response, missed decision or omission on the download through which fatigue could have acted. The hypothesis was not set aside; it was tested against the data and failed.

The condition still matters. The Stage 4 night roster puts every haul truck operator in charge of a 220 t class vehicle in the small hours of a tenth consecutive night. There is no fixed lighting anywhere in the Stage 4 pit, so the operator's own eyes are the only defence against an unlit hazard at an unlisted intersection. The same roster is worked on three other pits on the site. It is a standing exposure built into the roster, not a feature of one night.

The investigation found it virtually certain that the truck operator was working in the eleventh hour of the tenth consecutive night shift during the period of lowest circadian alertness, unlikely that this contributed to the collision, and likely that it increased risk.

Neither operator aware of the other

The factors examined throughout this analysis converge on one condition: two vehicles arriving at the same intersection at night, each unknown to the other.

The truck. HT-407 descended empty at 37 km/h with no intersection at RL 312 on its in-cab road network and no warning sign on the approach. There was no call on the channel, and no alert from a collision awareness system that had logged the light vehicle at 21.4 m. The operator knew the spur had been cut and had noticed the construction sign was gone, but believed the spur was still closed.

The light vehicle. LV-118 left the spur at about 25 km/h with 38 m of sight line, no give-way control, no call and no knowledge of the truck.

Supervision. The night shift supervisor did not know a light vehicle was in the pit at all. No permit, call-in or board existed by which he could have known.

The first sighting available to either driver was at 41 m, about four seconds before the truck reached the intersection, which is what the survey predicts for the geometry. The radio log, the download, the collision awareness system log and the three interviews contain no indication that either operator knew the other was there before that moment.

The light vehicle's movements before the final seconds rest on the driver's own account, which is the one reservation on this conclusion. On whether the condition led to the collision there is no real doubt. Nothing between the convergence and the impact could have prevented a collision with 41 m of sight line.

The investigation found it extremely likely that both vehicles arrived at the intersection with neither operator aware of the other, and extremely likely that this contributed to the collision.

Environment

The crest and the sight line at the RL 312 intersection

The survey of 13/08/2026 places the spur mouth 34 m beyond the high point of a crest in the ramp.

Observer	Eye height	First line of sight
Operator descending the ramp	5.80 m	41 m to the spur mouth
Driver at the spur mouth	1.45 m	38 m to a descending truck

The 1.6 m inner windrow hid the truck's lower body from the spur for the last 25 m of its approach.

BIN-TMP-004 clause 6.2 requires that "at every intersection, the available sight distance shall be not less than the stopping distance of the largest vehicle using the road at the posted speed, and in no case less than 120 m. Where this cannot be achieved, the intersection shall be controlled by a stop control, signalised control or traffic controller." The survey calculates the stopping distance of a 220 t class truck travelling empty at the posted 40 km/h on a 9 per cent descending grade at 96 m, using the manufacturer's data and a 1.5 second response. The available sight distance was 41 m. As the survey concludes, the intersection "does not meet clause 6.2 on either limb". It had none of the controls clause 6.2 requires where sight distance cannot be achieved.

The consequence is direct. With 41 m of sight line against 96 m of stopping distance, a truck travelling at the posted speed could not stop for anything at this intersection, however alert the operator was or however quickly he reacted. The geometry gave about four seconds between first sighting and arrival. Attention, training and a radio call all operated within limits that the geometry had already set.

The intersection also carried none of the controls that clause 6.4 requires at intersections on the road layout drawing:

- no warning sign on either ramp approach
- no give-way or stop control at the spur mouth
- no delineation, reflective marker posts or windrow gap markers
- no posted speed for the spur.

A ROAD CLOSED - CONSTRUCTION sign had stood at the spur mouth while the road was cut. It was removed on 02/08/2026, the day the bore was energised and light vehicles first used the spur, and nothing replaced it.

The hazard was reported in writing seven days before the occurrence. After a near miss at the same intersection, the dewatering technician who submitted hazard report HAZ-2026-0917 wrote: "The problem is you cannot see up the ramp from that road. There is a rise in the ramp and a windrow along the side and you are out on the road before you can see anything." He added that "There is no sign on the ramp telling the trucks the road is there and there is no give way on our side," and that "The trucks are doing forty coming down empty. I do not think they know the road is there." How that report was handled is analysed under Procedures.

A sight line is dangerous only once something is in it. It was the other conditions analysed in this report that placed two vehicles in it at the same moment. That is the only reservation on the influence of the geometry.

The investigation found it virtually certain that the spur met the ramp 34 m beyond a crest, giving 41 m of sight line against a stopping distance of 96 m and a plan minimum of 120 m, and very likely that this contributed to the collision.

Darkness and lighting

The collision occurred at 02:47 in darkness. Moonset had been at 23:18 the previous evening. There is no fixed lighting anywhere in the Stage 4 pit. The only light came from vehicles and from the light plants at the loading face, 1.1 km away and out of sight behind the crest. The light vehicle's beacon was on, and its lamp worked when tested after recovery.

These conditions form the setting for the sight line and dust discussions. They were not tested as a separate factor, and no finding is made on them.

Dust on the ramp: an unresolved matter

The investigation examined whether dust on the ramp reduced the truck operator's ability to see the light vehicle. The question could not be resolved. The investigation assessed it as about as likely as not that dust affecting visibility was present, and it makes no finding either way.

Evidence that dust may have been present

- The water cart log records the section of ramp between RL 280 and RL 400 last watered at 01:10, 1 hour 37 minutes before the occurrence. The standing night shift instruction is a pass every 90 minutes, so a pass was 7 minutes overdue.
- The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre in the 02:40 to 02:50 interval, against a shift mean of 88.
- The truck operator described dust hanging on the ramp and said the cart had been through earlier and the surface dries out.
- The truck was descending behind its own lights into an unlit pit, which is the condition in which airborne dust is worst for a driver.

Evidence against

- The light vehicle driver, who was at the same place four seconds earlier, was firm that the air was clear, that he could see the ramp surface and the windrow, and that dust at Bindarra is nothing like that at the other sites where he works.
- The truck operator declined to say that the dust made a difference and asked not to be recorded as saying so.
- The reading of 142 was well below the site trigger of 250. It measures airborne particulate at a monitor 180 m away, not what could be seen from either cab.
- The operator first saw the light vehicle at about 40 m. The survey gives 41 m for perfectly clear air, so his account and the sight line agree without any role for dust. This is the strongest point against the hypothesis: dust is not needed to explain anything that happened.

Why it could not be resolved

The departure from normal conditions was real, at the right place and at the right time. Its effect on detection is not established either way. Nothing at Bindarra measures visibility, and there is no measurement of what could be seen from either cab on the night. The two accounts are both

credible and do not agree, and the one measurement available shows higher particulate levels but not their effect on vision. Only a measurement of visibility from the cabs at the time could settle the question, and none exists.

The matter is unresolved and is not carried into the findings.

Equipment

Collision awareness alerts suppressed at the intersection

HT-407's collision awareness system operates at Level 7 (operator advisory). Level 9 (automated intervention) is available on the fleet but is not enabled at Bindarra. The system raises an advisory alert when a vehicle tag enters the 60 m detection ring at a closing rate above 10 km/h, except inside a suppression geofence.

The S4-LOAD geofence was drawn on 14/03/2025 around the Stage 4 loading face. Its purpose was to stop nuisance alerts between the excavator, the trucks under it and the service vehicles working nearby, where the alert rate had made the system unusable. On 09/06/2026, when the face advanced, the contractor's technical services group extended the geofence northward as a system configuration task. The new boundary lies 180 m up-ramp of the RL 312 intersection, so the intersection sat inside the suppressed area. The extension is not recorded in the traffic management plan and was not reviewed against it. No training record anywhere on site, for any person, covers the suppression geofences.

At 02:47:09.2 the system logged a proximity event with tag LV-118 at 21.4 m, closing at 13 km/h. The event log reads "alert SUPPRESSED (geofence)". Outside the geofence the same event would have raised an advisory tone and a screen indication in the cab.

The system's own twelve-month statistics show a pattern:

Period: 12 months to the occurrence	Count
Advisory alerts raised by HT-407	412, every one outside the geofence, none inside it
Proximity events suppressed inside the geofence, Stage 4 haul fleet	68
Of those, a light vehicle tag on the main ramp below the boundary	9

The nine events involved the same combination of vehicles and location as the occurrence. They were recorded and visible to anyone who queried the system, but nobody had queried it. The technical services group's standing report covers alert counts and system health and does not report suppressed events at all.

The evidence on whether the suppression contributed supports two readings, and both are set out here.

Against a contribution. On this occasion the alert would have sounded at 02:47:09.2. That was 1.8 seconds after the operator had begun braking and 2.8 seconds before impact, so an advisory tone would have told him something he already knew. The system is advisory only and has no intervention function. With 41 m of sight line, the outcome was fixed before either the alert or the sighting. On this reading the suppression is a serious matter for the fleet but not a cause of this collision.

For a contribution. The influence lies not in the alert that did not sound in the final three seconds but in the expectation the system had built up beforehand. On that ramp the system raised no alerts inside the geofence, and every alert HT-407 raised in the preceding year came from outside it. In his own account, and before he was asked about it, the operator said he had come to treat a

quiet system as an empty road. The night shift supervisor also raised the suppression unprompted, and the statistics support both accounts. That expectation was a condition the operator carried over the crest, and it is one of the reasons he was unaware of the light vehicle before sighting it.

The investigation accepted the second reading on the balance of the evidence. The competing reading is strong enough that this conclusion is less secure than the findings on the geometry and the radio call.

The finding does not concern the existence of the geofence: nuisance alerts around an excavator can make a collision awareness system unusable. It concerns the boundary. The geofence was moved 180 m in June 2026 into a section of haul road that soon afterwards acquired an intersection, and nobody outside the group that moved it was told.

The investigation found it virtually certain that the collision awareness alerts were suppressed at the RL 312 intersection by a geofence extended to cover it, and likely that this contributed to the collision.

The in-cab road network

The fleet management system in-cab map for the Stage 4 pit was last published on 24/07/2026. The spur was cut on 31/07/2026, and on the night of the occurrence the in-cab display showed the main ramp with no intersection at RL 312. Publishing a revised road network is a survey task triggered by a management of change record, and no such record existed for the spur. The absence of the spur from the map is part of the works order finding under Organisation, and no separate finding is made here.

The impact

The impact occurred at 02:47:12.0 at 24 km/h, 4.6 seconds after brake application. It was 6.2 m into the ramp from the eastern edge of the spur mouth and 3.1 m out from the inner windrow toe. The truck's accelerometer logged 1.9 g longitudinal. Four sources agree on the configuration: the recorded data, the damage and the debris field.

- Damage to LV-118 was concentrated on the right side from the B pillar forward.
- Damage to HT-407 was confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line.
- Right-side glass and mirror fragments lay from 4.8 m to 14 m down-ramp of the impact position.

The truck struck the side of the light vehicle, not its front. That is the configuration the intersection geometry makes almost certain: a light vehicle emerging across the path of a descending truck presents its side. A 220 t class vehicle striking the side of a light vehicle leaves the outcome to three things: the light vehicle's structure, the restraint and the impact speed. The structure and the restraint were fixed before the sighting. The speed was the one thing still open, and the truck operator's braking reduced it.

The investigation found it virtually certain that the haul truck's front left struck the right side of the light vehicle at 24 km/h, and virtually certain that this contributed to the driver's injuries.

The roll-over, and the defences that held

A continuous gouge runs 18.4 m down-ramp from the impact position. At 11.2 m it changes from tyre scuff to body panel gouge, which marks where the vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp. Its door was against the ground, so the driver was extricated through the rear.

The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. These are the injuries a side impact and roll-over produce when the structure holds and the restraint is worn, and both defences did hold:

- The certified roll-over protection structure was deformed inward at the right rear mounting but was not breached.
- The driver's seatbelt was found fastened, with loading marks on the webbing.

Both defences were loaded to the point of showing it. Without either of them, the same sequence would very probably have had a more serious outcome. The light vehicle had been serviced on 17/07/2026. Its inspection sheet for the week of 09/08/2026 recorded all items serviceable.

The investigation found it virtually certain that the light vehicle was pushed 18.4 m down the ramp and rolled onto its left side, and virtually certain that this contributed to the driver's injuries.

Procedures

Positive calls required only at listed intersections

BIN-TMP-004 is the mine's own defence against the kind of interaction that occurred. Clause 4.3.1 requires that "A positive call shall be made on the relevant pit channel before entering any intersection listed at Appendix B, and entry shall not be made until the call is acknowledged or the channel is confirmed clear." Clause 4.3.2 requires no call at any other intersection. Appendix B lists nine intersections, and a laminated copy is posted in each crib room. The board in the Stage 4 crib room was the 12/11/2025 issue.

The control is sound in itself. The route by which an intersection reaches it is circular. The note to Appendix B provides that the appendix "is reviewed and reissued at the annual review of this plan (clause 9.1)", and that an intersection is added "where it appears on the road layout drawing at Schedule 1 and the review assesses it as requiring positive communication." Schedule 1 is dated 12/11/2025 and is reissued only at the annual review. The plan contains no provision for adding an intersection to either Appendix B or Schedule 1 between annual reviews, and no requirement that a new road be referred to the plan's owner when it is cut. Clause 9.1 does provide for review "following any significant change to the mine's road network", but no out-of-cycle review had been held in the plan's eleven-year history.

The RL 312 spur appeared on neither document. The earliest date on which it could have been listed was the next annual review, due on 12/11/2026, three months after the collision.

The pattern extends beyond this intersection:

- **Unreferred roads.** The survey department's road register records 14 new roads cut in the Stage 4 pit since the 12/11/2025 review. The three built under a management of change record were referred to the plan's owner; none of the eleven built under a works order was.
- **Hazard reports.** A search of the Corella Resources reporting system for the 24 months before the occurrence returned four hazard reports, including HAZ-2026-0917. Each of the other three, from 03/2025, 09/2025 and 02/2026, described a light vehicle and a haul truck meeting unexpectedly at an intersection not listed at Appendix B, at three different locations.
- **Hazard report closure.** All four reports were closed within three days by the department that received them, with an entry recording that the crew had been reminded to take care and that heavy vehicles have right of way. None was referred to the plan's owner, and none resulted in a change to Appendix B, Schedule 1 or any physical control.

HAZ-2026-0917 followed the same path. The Dewatering Superintendent's closure entry records that the crew had been reminded "to take care entering the main ramp and to give way to haul trucks at all times", and concludes: "No further action required. Closed." The risk rating was reduced from Medium to Low.

Had a call been required at this intersection, it would very probably have made a difference. The channel was open, both vehicles were on it, and the dispatcher was already placing the truck to within minutes. The reservation is that a call made but not acknowledged still leaves a driver deciding whether to go.

The investigation found it virtually certain that the traffic management plan required positive calls only at listed intersections, and provided for new intersections to be added to that list only at the annual review, and very likely that this contributed to the collision.

The external guidance and the discretion it leaves

BIN-TMP-004 is made to satisfy Mines Safety Guideline MSG-14 revision 4, issued by the State Mines Inspectorate under regulation 6.22 of the Mining Operations Safety Regulations 2018. The regulation itself requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine". The guideline leaves three key terms to the site to define:

- **Clause 3.2** requires the plan to be reviewed "at intervals not exceeding twelve months, and following any significant change to the mine's road network". It does not define "significant change" or give an example of one.
- **Clause 5.1** requires the operator to nominate the intersections at which positive communication is required. It does not say on what basis they are nominated and sets no minimum.
- **Clause 5.4** allows detectability of a light vehicle to be provided "by line of sight, elevated marking, radio communication, a proximity detection system, or a combination of these". It does not say which, or how many.

Under clause 3.5 the plan "is a site document. It is not approved by, and need not be submitted to, the Inspectorate, except on request following an occurrence or an inspection."

On the day of the occurrence Bindarra's plan met the guideline on its face. It had been reviewed within twelve months, it nominated nine interaction zones, and it provided for detectability by line of sight and by elevated marking. The mine was not in breach. This finding concerns the shape of the discretion the guideline leaves, not any failure by the Inspectorate or by the mine to comply.

For influence. At Bindarra all three open terms were filled in by the same document, reviewed once a year, with no outside review of any part of it. A defined review trigger and a stated minimum for nominated intersections would have given the eleven unreferred roads a route into the plan. Every internal control examined in this occurrence failed for want of exactly that route.

Against influence. The guideline did not prevent anything. The site could have held an out-of-cycle review under its own clause 9.1 at any time, and MSG-14 does not stop a mine from listing every intersection it has. On this reading Bindarra did not fail to act because the guideline was vague. It failed to act because nothing inside the mine ever put the question to the plan's owner, and the external guidance is background.

The investigation accepted the case for influence on the balance of the evidence. The argument against it is substantial, and this conclusion is less secure than the other findings on how the occurrence came about.

The investigation found it virtually certain that the traffic management guidance left the review trigger undefined and the nomination of interaction zones to the site without a minimum or basis, and likely that this contributed to the collision.

Organisation

The spur built under a works order, with no management of change

CRL-PRO-022 clause 5.2 requires that "A management of change record shall be raised before any of the following: a change to the mine's road network; the introduction of a new vehicle type; a change to a pit design; a change to a documented control." Clause 5.3 excludes "routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced."

Clause 5.4 requires the change owner, once a record is raised, to complete an impact assessment. That assessment includes referral to three parties:

- the traffic management plan owner
- the survey department, for the fleet management system road network
- the contractor's technical services group, for the collision awareness system configuration.

Those three referrals exist only inside the management of change record. The works order form at Schedule 4 has no equivalent fields and no referral of any kind.

Works order WO-26-4471 was raised by the Mining Engineer on 28/07/2026 and approved by the Mining Superintendent on 29/07/2026. Its scope was to "Cut and form 240 m of access road to dewatering bore DW-14 at RL 312". It classified the work as "Routine earthworks within the approved Stage 4 design. No new plant or vehicle type. Management of change not required, CRL-PRO-022 clause 5.3." The works were completed on 31/07/2026. The handover field reads "N/A". The works order contains no scope for signage, survey pickup of the new intersection, referral to the traffic management plan, update to the road network, or review of the geofences.

The classification decided everything that followed. None of the three referrals was made:

- The plan's owner was not told, so the spur never reached Schedule 1 or Appendix B.
- The survey department was not asked to publish a revised road network, so the in-cab map published on 24/07/2026 remained current on the night and showed no intersection at RL 312.
- The technical services group was not asked to review the geofences, so nobody compared the new intersection with the S4-LOAD boundary moved two months earlier.

Because the intersection was not on Schedule 1, clause 6.4 did not apply to it, and it received no signage, give-way control or delineation. Because clause 6.2 was never applied either, nobody measured its sight distance until the investigation asked for a survey.

The spur was not an isolated case. Eleven of the fourteen new Stage 4 roads cut since the last plan review were built in the same way.

A management of change record was the only route by which the new intersection could have reached the plan, the in-cab map or the geofence configuration. Its absence left all three unchanged. The reservation is that a referral is not the same as a control: the plan's owner might have considered the spur and still left it unlisted.

The investigation found it virtually certain that the spur was cut and opened to traffic under a works order with no management of change record raised, and very likely that this contributed to the collision.

Verification of the light vehicle driver's site induction

CRL-STD-009 clause 7.3 allows a contractor's own general induction to be accepted as equivalent where the contractor holds a current recognition agreement. Merrivale Survey & Geotechnical has held one since 2023. Clause 7.4 sets a clear limit on that equivalence: it "does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive a vehicle in an operating pit and shall not be substituted." Clause 7.5 makes the engaging department responsible for verifying that 7.4 has been completed before a site access card is issued.

The standard says the right thing. The failure lay in where the verification was supposed to happen. The gatehouse issues cards against the recognition agreement alone. It holds no record of the supplement or the pit driving assessment and is not asked for one. Nothing in the access card system distinguishes a person who may drive in an operating pit from one who may not.

The driver's training file records:

- a two-hour Merrivale online general mining induction, completed 24/07/2026
- a 20 minute gatehouse briefing covering muster points, speed limits and personal protective equipment
- a site access card issued on 09/08/2026.

There is no record of the site-specific supplement or the pit driving assessment. The driver had never seen BIN-TMP-004 and did not know it existed. He was driving a light vehicle on an active haul road on his third night at the mine without the assessment the site's own standard says may not be substituted. No part of the system that admitted him was checking for it. This is a finding about that system, not about the driver.

The pit driving assessment is where a light vehicle driver learns how the pit's traffic rules work in practice: give-way, positive calls, and what to do at a blind intersection. Two considerations limit how much it could have changed. The rule he would have been taught, that light vehicles give way at all times, is one he was already applying. And the intersection's sight line defeated attention on its own, as analysed under Environment. The records for the site supplement are held by the driver's employer and were not available to settle its status beyond the gatehouse register and training file.

The investigation found it extremely likely that the contractor's induction was accepted as equivalent without the site supplement or the pit driving assessment being verified, and likely that this contributed to the collision.

Night shift emergency response capability

The Bindarra night shift is rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder, who must be present before any extrication begins, is called in from the accommodation village 11 km from the gatehouse. On the night:

- Dispatch paged the team at 02:48.
- The first response vehicle entered the pit at 02:51 and reached the scene at 02:54.
- The second responder arrived with the hydraulic rescue tool at 03:07.
- The driver was extricated at 03:14, 27 minutes after the collision.

The site's post-incident review found that extrication was delayed by approximately 9 minutes waiting for the second responder and the tool. It recorded that the delay did not affect the clinical outcome, because the injuries were stable and the driver was conscious throughout.

This arrangement did not influence the occurrence or its outcome. The collision and the injuries were complete before the emergency response was paged, and the post-incident review confirms on clinical grounds that the delay cost the driver nothing.

It remains a significant gap in a last line of defence. The driver was conscious, stable, restrained and protected by an intact roll-over structure. With a fuel fire, an unconscious casualty or an airway requiring the vehicle to be opened within the first ten minutes, the same nine minutes could produce a very different outcome. The gap is structural rather than clinical. On night shift the site's extrication capability is one person plus a drive from the village, on every shift, and nothing in the arrangement changes with the severity of the casualty.

The investigation found it virtually certain that the night shift was rostered with one paramedic and a second responder called from the village, unlikely that this contributed to the occurrence, and very likely that it increased risk.

Hypotheses tested and excluded

Two hypotheses were tested and ruled out.

A wet ramp surface. The first was that the ramp surface was wet and the truck could not stop in the distance available. The investigation assessed this as very unlikely. The met station recorded no precipitation, and the last rainfall at the site had been 26 days before. The last water cart pass on the section was 1 hour 37 minutes before the occurrence, longer than the standing 90 minute interval. The investigator examined the ramp at 06:20 and found it dry throughout, with no damp patches, ponding or greasy film at or above the intersection. The 62 m braked scuff showed no sign of a locked or sliding wheel, and the deceleration achieved matched the unit's tested dry figure.

Degraded service brakes. The second was that the truck's service brakes were degraded. The investigation assessed this as very unlikely. The brake performance tests on 03/08/2026 and 14/08/2026 gave 1.9 m/s^2 and 1.92 m/s^2 , and the occurrence download gave 1.85 m/s^2 , between the two tests and within normal scatter. Brake pressure rose normally and the retarder was at maximum throughout. No brake, retarder or park brake defect had been raised in the preceding twelve months, and the pre-start check for 11/08/2026 recorded no defects.

Findings summary

These findings are the conclusions the investigation reached about the collision between a haul truck and a light vehicle at the RL 312 spur intersection at Bindarra Gold Mine on 11/08/2026. They should not be read as apportioning blame or liability to any organisation or person.

Contributing factors

- The collision awareness alerts on HT-407 were, with virtual certainty, suppressed inside a geofence that had been extended to cover the new intersection, and this suppression likely contributed to the occurrence. (Safety Issue)
- It was very likely that both vehicles arrived at the intersection with neither operator aware of the other, and very likely that this contributed to the collision.
- The haul truck's front left struck the right side of the light vehicle at 24 km/h, and this impact contributed to the occurrence.
- With virtual certainty, the light vehicle was pushed 18 m down the ramp and rolled onto its left side, and with virtual certainty this roll-over contributed to the serious injuries the driver sustained.
- Traffic management guidance left the review trigger undefined and the interaction zones to the site. This existed with virtual certainty and likely contributed, because it gave no defined point at which a new road had to be brought into the site's traffic management plan. (Safety Issue)
- The access spur was cut and opened to traffic under a works order with no management of change raised, which existed with virtual certainty and very likely contributed, because the new intersection never reached the traffic management plan, the in-cab road network or the collision awareness system configuration. (Safety Issue)
- Surveyed with virtual certainty, the spur met the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m, and this geometry very likely contributed to the collision.
- Under BIN-TMP-004, positive radio calls were required only at listed intersections, and the list was rewritten once a year. This existed with virtual certainty and very likely contributed, because the RL 312 intersection could not be listed until the next annual review. (Safety Issue)
- Neither operator made a positive call on the pit channel before entering the intersection, which existed with virtual certainty and likely contributed, because neither vehicle was placed in the other's knowledge.
- A contractor induction was very likely accepted as equivalent, with the site supplement and pit driving assessment never verified, and on balance this contributed to the way the light vehicle entered the ramp. (Safety Issue)
- The light vehicle entered the ramp from the spur without stopping and without establishing the truck's position, which existed with virtual certainty and very likely contributed to the collision.

Other factors that increased risk

- Night shift emergency response at Bindarra was, with virtual certainty, provided by one paramedic rostered on night shift, with the second responder called from the village 11 km from the gatehouse, and this arrangement very likely increased risk to anyone requiring extrication at night. (Safety Issue)

- With virtual certainty, the haul truck operator was on night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low, and this roster exposure likely increased risk.

Other findings

- The hypothesis that the ramp surface was wet and the truck could not stop in the distance available was examined and ruled out.
- The hypothesis that the haul truck's service brakes were degraded and could not achieve the required stopping performance was examined and ruled out.
- The truck operator braked and steered into the windrow within 1.4 seconds of sighting, and this response reduced risk.

Appendices

Appendix A - Summary of factor testing

Each factor the investigation examined was tested in three stages:

- whether it existed
- whether it influenced the occurrence
- whether it was important enough to warrant action even where it did not influence the occurrence

The results below use the likelihood terms applied when each factor was tested.

Factor	Lane	Existence	Influence	Importance	Category
Traffic management guidance leaves the review trigger undefined and the interaction zones to the site	External Influences	Pass - virtually certain	Pass - likely	Not tested	Contributing factor (safety issue)
Access spur cut and opened to traffic under a works order, with no management of change raised	Organisational Influences	Pass - virtually certain	Pass - very likely	Not tested	Contributing factor (safety issue)
Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified	Organisational Influences	Pass - extremely likely	Pass - likely	Not tested	Contributing factor (safety issue)
Positive radio calls required only at listed intersections, and the list is rewritten once a year	Risk Controls	Pass - virtually certain	Pass - very likely	Not tested	Contributing factor (safety issue)
Collision awareness alerts suppressed inside a geofence extended to cover the new intersection	Risk Controls	Pass - virtually certain	Pass - likely	Not tested	Contributing factor (safety issue)
The spur meets the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m	Local Conditions	Pass - virtually certain	Pass - very likely	Not tested	Contributing factor
Neither operator made a positive call on the pit channel before entering the intersection	Individual Actions	Pass - virtually certain	Pass - likely	Not tested	Contributing factor
Light vehicle entered the ramp from the spur without stopping and without establishing the truck's position	Individual Actions	Pass - virtually certain	Pass - very likely	Not tested	Contributing factor
Both vehicles arrived at the intersection with neither	Occurrence Events	Pass - extremely likely	Pass - extremely likely	Not tested	Contributing factor

operator aware of the other					
The haul truck's front left struck the right side of the light vehicle at 24 km/h	Occurrence Events	Pass - virtually certain	Pass - virtually certain	Not tested	Contributing factor
The light vehicle was pushed 18 m down the ramp and rolled onto its left side	Occurrence Events	Pass - virtually certain	Pass - virtually certain	Not tested	Contributing factor
One emergency response paramedic rostered on night shift, with the second responder called from the village	Risk Controls	Pass - virtually certain	Fail - unlikely	Pass - very likely	Other factor that increased risk (safety issue)
Night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low	Local Conditions	Pass - virtually certain	Fail - unlikely	Pass - likely	Other factor that increased risk
Truck operator braked and steered into the windrow within 1.4 seconds of sighting	Individual Actions	Pass - virtually certain	Pass - very likely	Not tested	Positive event (reduced risk)
Dust on the ramp reduced the truck operator's ability to see the light vehicle	Local Conditions	Inconclusive - about as likely as not	Not tested	Not tested	Unresolved; no finding made
The ramp surface was wet and the truck could not stop in the distance available	Local Conditions	Fail - very unlikely	Not tested	Not tested	Excluded
The haul truck's service brakes were degraded and could not achieve the required stopping performance	Occurrence Events	Fail - very unlikely	Not tested	Not tested	Excluded

Notes on the table

The influence result for the traffic management guidance was argued, and accepted on balance. That result carries less confidence than the other influence results in the table.

The influence result for the collision awareness alert suppression was also argued. Two readings were considered:

- the fourteen months without an alert on the ramp set the operator's expectation that silence meant an empty road
- on this occasion the alert would have come after braking had already begun

The dust factor was examined and could not be resolved either way. It is not a finding. The reasons are set out in the Analysis.

Appendix B - Excluded hypotheses

The ramp surface was wet and the truck could not stop in the distance available

This hypothesis was tested because an 18 m push down a 9 per cent grade raises a question about the road surface. It was found very unlikely that such a surface existed. Three independent sources agreed:

- **Weather:** the met station recorded no precipitation on the night. The last rainfall at the site was 26 days before the occurrence.
- **Watering:** the water cart log put the last pass on that section at 17:10 UTC, 1 hour 37 minutes before the occurrence. The standing night shift instruction was a pass every 90 minutes, so the surface had been drying for longer than the watering interval rather than being freshly watered.
- **Surface condition:** the investigator examined the ramp at 06:20 local. It was dry throughout, with no damp patches, ponding or greasy film at or above the intersection.
- **Braking evidence:** the braked tyre scuff ran continuously for 62 m with no sign of a locked or sliding wheel. The fleet management system download gave a mean deceleration of 1.85 m/s² against the unit's own tested 1.9 m/s² on a dry 9 per cent grade.

No source gave any evidence of a wet or slippery surface.

The haul truck's service brakes were degraded and could not achieve the required stopping performance

This hypothesis was tested because a 220 t truck that does not stop in the distance available first raises the question of its brakes. It was found very unlikely that such a degradation existed. The braking system performed to specification before, during and after the occurrence:

- **Test before the occurrence:** the monthly brake performance test on 03/08/2026, eight days before, gave a mean deceleration of 1.9 m/s² empty on a 9 per cent grade.
- **Test after the occurrence:** the test repeated on 14/08/2026, before the unit returned to service, gave 1.92 m/s².
- **The occurrence itself:** the download gave 1.85 m/s² over the 4.6 seconds of braking. This sits between the two test results and within normal scatter.
- **Brake response:** service brake pressure rose from 0 to 620 kPa in 0.4 seconds, and the retarder was at maximum throughout.
- **Defect history:** no brake, retarder or park brake defect was raised against the unit in the twelve months before the occurrence. The pre-start check sheet for 11/08/2026 was complete and recorded no defects.

Appendix C - Sources of evidence

Interviews

Role	Date	Method
Haul truck operator	Completed by 21/08/2026 [NOT ESTABLISHED: date of this interview]	Recorded and transcribed
Night shift supervisor	Completed by 21/08/2026 [NOT ESTABLISHED: date of this interview]	Written up and read back to the interviewee
Light vehicle driver	Completed by 21/08/2026 [NOT ESTABLISHED: date of this interview]	Written up and read back to the interviewee. The driver corrected his account on read-back.

Documents, data and physical evidence

Material	Reference and date
Occurrence notification	CRL-OCC-2026-0332, raised 11/08/2026 19:26 UTC
Scene examination, RL 312 intersection, including signage and delineation	INV-2026-0332-SCENE, examined 12/08/2026 before recovery

Fleet management system download, HT-407	Warrigal Mining Services technical services report WMS-TS-2026-114, issued 18/08/2026
Collision awareness system configuration and event log, HT-407	Extracted 12/08/2026, WMS-TS-2026-114
Collision awareness system alert statistics	Warrigal technical services fleet CAS event log extract, 12 months to 11/08/2026
Pit channel radio recording, 18:20 to 18:52 UTC	RAD-2026-0332, transcribed 17/08/2026 and verified by a second investigator
Survey of the RL 312 intersection and sight distances	Mine survey pickup SUR-26-1188, 13/08/2026
Hazard report and closure entry	HAZ-2026-0917, submitted 04/08/2026, closed 06/08/2026
Works order, RL 312 dewatering access	WO-26-4471, raised 28/07/2026, completed 31/07/2026
Bindarra Traffic Management Plan, clauses 4, 6 and 9 and Appendix B	BIN-TMP-004 revision 7, issued 12/11/2025
Traffic management plan review history and hazard report extract	Document register BIN-TMP-004; Corella reporting system search, 24 months to 11/08/2026
Management of Change procedure, clauses 5.2 to 5.4 and Schedule 4	CRL-PRO-022 revision 3
Contractor Management Standard, clauses 7.3 to 7.5	CRL-STD-009 revision 5
Mines Safety Guideline, clauses 3.2, 3.5, 5.1 and 5.4	MSG-14 revision 4, made under Mining Operations Safety Regulations 2018 r 6.22
Training, competency and induction records, both drivers	Warrigal training file, Merrivale training file and the Bindarra gatehouse access card register
Haul truck operator roster, duty and sleep record	Warrigal roster and site access records; the operator's own sleep record, provided voluntarily
Light, surface, weather and dust records	Met station, water cart log and fixed dust monitor records, 11/08/2026
Equipment maintenance, inspection and brake test records	Warrigal asset register and Corella light vehicle register, obtained 13/08/2026
Emergency response log and post-incident review	Obtained 26/08/2026
Drug and alcohol testing, haul truck operator and shift supervisor	Conducted on the night of the occurrence; both negative

Appendix D - Investigation activity

Times in this appendix are UTC.

11/08/2026, 20:15: The investigator was notified by the Night Shift Superintendent. The occurrence was assessed as a reportable serious injury warranting investigation. The investigation was opened, its scope set, and statutory notification made to the State Mines Inspectorate.

12/08/2026:

- The scene was attended before recovery. Both vehicles were photographed and measured in position, and the marks were recorded.
- The fleet management and collision awareness system downloads were taken by the contractor and handed over sealed.
- Both operators were tested for drugs and alcohol, with negative results.

- A site-wide direction was issued overnight requiring a positive call at every Stage 4 intersection, listed or not.

13/08/2026:

- A survey pickup of the intersection was requested and received. It gave 41 m of available sight distance against a plan minimum of 120 m and a stopping distance of 96 m.
- Works order WO-26-4471 was located. It recorded management of change as not required, and its handover field read N/A.
- Hazard report HAZ-2026-0917 and its closure entry were obtained from the reporting system.

21/08/2026: Interviews with the haul truck operator, the night shift supervisor and the light vehicle driver were completed. Collision awareness system alert statistics were requested following the accounts of the operator and the supervisor.

28/08/2026: A preliminary update was issued.

01/09/2026: The investigation confirmed that there was no route to add an intersection to Appendix B of BIN-TMP-004 between reviews. An intersection reaches the list only through a drawing reissued once a year, and nothing requires a new road to be referred to the plan's owner. The matter was raised with the Mining Manager.

Distribution

Title	Name
General Manager, Bindarra Gold Mine	T. Brandner
Registered Manager (statutory)	S. Kirkhope
Mining Manager and owner of BIN-TMP-004	H. Quaile
Manager Health and Safety, Corella Resources	L. Farquhar
Project Manager, Warrigal Mining Services	D. Mokoena
Superintendent Dewatering, Corella Resources	P. Nyaguse